MPM-Geomechanics

An open-source simulator for large-deformation geomechanics



Technical Manual

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November 6, 2026

1 Introduction to the Material Point Method (MPM)

The Material Point Method is an hybrid Lagrangian-Eulerian numerical method, that allows for simulating continuum mechanics processes involving large deformations and displacements without issues related to computational mesh distortion. In MPM, the material domain to be simulated is discretized into a set of material points that can move freely within a computational mesh, where the equations of motion are solved. The material points store all variables of interest during the simulation, such as stress, pore pressure, temperature, etc., giving the method its Lagrangian characteristic.

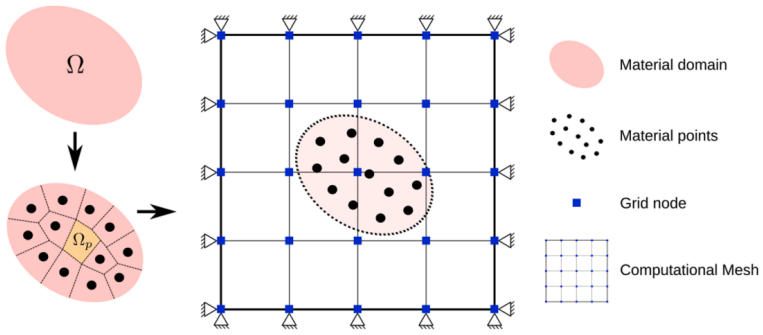


Figure 1: General MPM approach. Solid and space discretization with Lagrangian material points and structured Eulerian mesh.

In an MPM computational cycle, all variables stored in the material points are computed at the computational mesh nodes using interpolation functions, and then the equation of motion is solved at the nodes. The nodal solution obtained is interpolated back to the particles, whose positions are updated, and all nodal variables are discarded. This method, enables the numerical solution of the motion equation in continuum mechanics by using the nodes of an Eulerian mesh for integration and Lagrangian material points to transfer and store the properties of the medium.

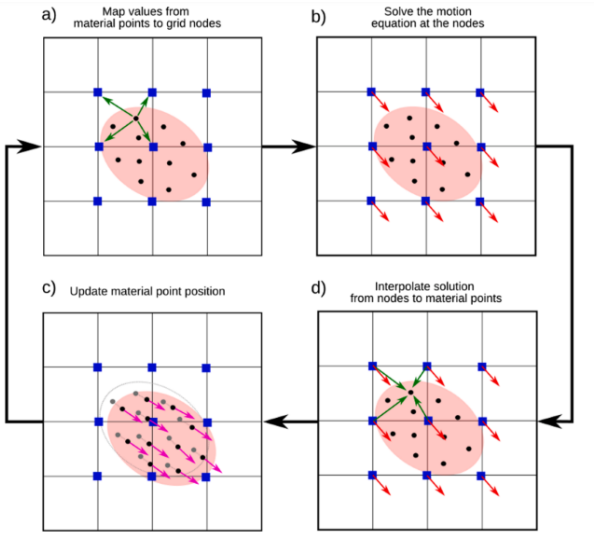


Figure 2: MPM computational cycle

2 MPM Formulation

The material point method formulation is based on the continuum mechanics motion equation in 3D:

$$\frac{\partial \sigma_{ij}}{\partial x_j} + \rho b_i = \rho a_i \tag{1}$$

The internal forces are related with the σ_{ij}, the Cauchy stress tensor, ρ is the mass density, b_i is a body force and a_i is the acceleration.

The equation **??** is presented here in tensor notation, but, vector and matrix can be equally used. The discrete form of the motion equation can be obtained using the weak form of this partial differential equation. The weak form is obtained by multiplying the motion equation by arbitrary weighting functions and integrating this product over the domain. In this procedure, the integration by parts reduces the order the stress tensor and introduces the natural boundary conditions:

$$-\int_\Omega \sigma_{ij}\delta u_{i,j} dV + \int_\Gamma t_i \delta u_i dA + \int_\Omega \rho b_i \delta u_i dV = \int_\Omega \rho a_i \delta u_i dV \tag{2}$$

Here δu_i are arbitrary displacements functions, whose value in the boundary are $\delta u_i|_\Gamma = 0$ and t_i is an external traction acting on the boundary Γ.

In the MPM context any field or space property $f(x_i)$ can approximated using the value stored in the particle f_p:

$$f(x_i) = \sum f_p \chi_p(x_i)$$

where χ_p is the particle characteristic function that defines the volume occupied by the material point:

$$V_p = \int_{\Omega_p \cap \Omega} \chi_p(x_i) dV$$

In consequent, density, acceleration and stress fields can be approximated by the values stored in particles:

$$\rho(x_i) = \sum_p \frac{m_p}{V_{ip}} \chi_p(x_i)\rho(x_i)a_i(x_i) = \sum_p \frac{\dot{p_{ip}}}{V_p}\chi_p(x_i)\sigma_{ij}(x_i) = \sum_p \sigma_{ijp}\chi_p(x_i)$$

where $\dot{p_{ip}} = m_p \dot{v}_{ip} = m_p a_S ip = f_{ip}$ is the momentum variation in time that is equal to the total force, regarding the second Newton's law.

Replacing these fields in the weak form of the motion equation we have:

$$-\sum_p \int_{\Omega_p \cap \Omega} \sigma_{ijp}\chi_p \delta u_{i,j} dV + \int_\Gamma t_i \delta u_i dA + \sum_p \int_{\Omega_p \cap \Omega} \frac{m_p}{V_p} b_{ip}\chi_p \delta u_i dV = \sum_p \int_{\Omega_p \cap \Omega} \frac{\dot{p}_p}{V_p}\chi_p a_i dV$$

In the generalized interpolation material point method (GIMP), the resolution of this equation is carried out using a Petrov–Galerkin scheme where the characteristic functions $\chi_p(x_i)$ are the trial functions and the nodal interpolation functions $N_I(x_i)$ are the test functions.To arrive at this scheme, the virtual displacements are expressed using nodal interpolation functions:

$$\delta u_i = \sum_I N_{Ip} \delta u_{iI}$$

The trial and test functions are such that:

$$\sum_I N_I(x_i) = 1 \sum_p \chi_p(x_i) = 1$$

The resulting discrete form of the motion equation then is:

$$f_{iI}^{int} + f_{iI}^{ext} = \dot{p}_{iI}$$

where

$$p_{iI} = \sum_p S_{Ip} p_{Ip}$$

is the nodal momentum,

$$f_{iI}^{int} = -\sum_p \sigma_{ijp} S_{Ip,j} V_p$$

is the nodal internal force, and

$$f_{iI}^{ext} = \sum_p m_p S_{Ip} b_{ip} + \int_\Gamma t_i N_I(x_i) dA$$

is the external force at node I.

The function S_{Ip} and its gradients $S_{Ip,j}$ are the weighting functions of node I evaluated at the position of particle p.

The GIMP shape functions are defined by

$$S_{Ip} = \frac{1}{V_p} \int_{\Omega_p \cap \Omega} \chi_p(x_i) N_I(x_i) dV$$

and

$$S_{Ip,j} = \frac{1}{V_p} \int_{\Omega_p \cap \Omega} \chi_p(x_i) N_{I,j}(x_i) dV$$

These two functions are also a partition of the unity $\sum_I S_{Ip} = 1$.

The weighting function need to be integrated over the particle domain by choosing different characteristic functions and interpolation functions in a Petrov–Galerkin scheme. In the contiguous particle GIMP (cpGIMP) the characteristic function in defined as step function and the interpolation function is defined as linear function:

$$\chi_p(x) = \begin{cases} 1, & x \in \Omega_p, \\ 0, & x \notin \Omega_p. \end{cases}$$

$$N_I(x) = \begin{cases} 0, & |x - x_I| \geq L, \\ 1 + \dfrac{x - x_I}{L}, & -L < x - x_I \leq 0, \\ 1 - \dfrac{x - x_I}{L}, & 0 < x - x_I < L. \end{cases}$$

Where the integration is performed analytically within the particle domain.

$$S_{Ip} = \begin{cases} 0 & |\xi| \geq L + l_p \\ \left(L + l_p + \xi\right)^2 / 4 L l_p & -L - l_p < \xi \leq -L + l_p \\ 1 + \xi / L & -L + l_p < \xi \leq -l_p \\ 1 - \left(\xi^2 + l_p^2\right) / 2 L l_p & -l_p < \xi \leq l_p \\ 1 - \xi / L & l_p < \xi \leq L - l_p \\ \left(L + l_p - \xi\right)^2 / 4 L l_p & L - l_p < \xi \leq L + l_p \end{cases}$$

and

$$\nabla S_{Ip} = \begin{cases} 0 & |x_p - x_I| \geqslant L + l_p, \\ \frac{L+l_p+(x_p-x_I)}{2Ll_p} & -L - l_p < x_p - x_I \leqslant -L + l_p, \\ \frac{1}{L} & -L + l_p < x_p - x_I \leqslant -l_p, \\ -\frac{x_p-x_I}{Ll_p} & -l_p < x_p - x_I \leqslant l_p, \\ -\frac{1}{L} & l_p < x_p - x_I \leqslant L - l_p, \\ -\frac{L+l_p-(x_p-x_I)}{2Ll_p} & L - l_p < x_p - x_I \leqslant L + l_p. \end{cases}$$

In which $2lp$ is the particle domain, L is the mesh size in 1D, and ξ is the relative particle position to node. Weighting functions in 3D are obtained by the product of three one-dimensional weighting functions:

$$S_{Ip}(x_{ip}) = S_{Ip}(\xi)S_{Ip}(\eta)S_{Ip}(\zeta)$$

where $\xi = x_p - x_I, \eta = y_p - y_I$ and $\zeta = z_p - z_I$.

3 Explicit MPM integration

The discrete form of the motion equation needs to be integrated in time for obtaining the solution in time t^{n+1}. The displacement, the velocity and the acceleration at time $t^0, t^1, t^2, ..., t^n$ are knows, and the values at time t^{n+1} are required, namely the solution of the problem. The time integration can be done by explicit and implicit methods. In explicit method, the solution t^{n+1} is obtained only with the current information f($t^n, ..., t^0$). In implicit method the solution needs to solve a system due the solution is in function of the form f($t^{n+1}, ..., t^0$).

3.1 Central difference Method

In central difference method, the velocity at $t^{n+1/2}$ can be approximated as:

$$\dot{u}^{n+1/2} = (u^{n+1} - u^n)\Delta t$$

and, the acceleration in t^n can be approximated as:

$$\ddot{u}^n = (\dot{u}^{n+1/2} - \dot{u}^{n-1/2})\Delta t$$

and therefore, the required displacement at t^{n+1} can be calculated as: $u^{n+1} = u^n + \dot{u}^{n+1/2}\Delta t$, where the velocity is

$$\dot{u}^{n+1/2} = \dot{u}^{n-1/2} + \ddot{u}^n \Delta t$$

.

The motion equation in time t^n is $m\,\ddot{u}^n = f^n$, therefore the acceleration in time t^n is $\ddot{u}^n = f^n/m$. Using this acceleration equation in the $\dot{u}^{n+1/2}$ we have the velocity $\dot{u}^{n+1/2}$:

$$\dot{u}^{n+1/2} = \dot{u}^{n-1/2} + f^n/m\,\Delta t$$

4 Numerical implementation of central difference method

For one Δt, the updated position can be obtained as:

Algorithm 1 Explicit time integration

1: Compute forces f^n
2: Compute acceleration $\ddot{u}^n = f^n/m$
3: Update velocity $\dot{u}^{n+1/2} = \dot{u}^{n-1/2} + \ddot{u}^n \Delta t$
4: Update position $u^{n+1} = u^n + \dot{u}^{n+1/2} \Delta t$
5: $n \leftarrow n + 1$

5 Stability Requirement

The central difference method is explicit here and conditionally stable, so the time step must be less that a certain value for avoiding error amplification. For linear systems this critical time step value depends on the natural period of the system. For undamped linear systems the critical time step is: $\Delta t_{cr} = T_n/\pi$, where T_n is the smallest natural period of the system. For finite element method, the critical time step of the central difference method can be expressed as:

$$\Delta t_{cr} = \min_e(l^e/c)$$

Where l^e is the characteristic length of the element and c is the sound speed. This time step restriction implies that time step has to be limited such that a disturbance, a mechanical wave, can travel across the smallest characteristic element length withing a single time step.

This condition is known as CFL condition, or Courant-Friedrichs-Lewy condition. For linear elastic material the sound speed (compression P wave) is:

$$c = \sqrt{\frac{E(1-\nu)}{(1+\nu)(1-2\nu)\rho}}$$

In the MPM, the particles can has velocities in any time step, so the critical time speed can be written with this velocity plus:

$$\Delta t_{cr} = l^e/\max_p(c_p + |v_p|)$$

In a structured regular mesh, l^e is the grid cell dimension. And c_p is the sound speed calculated with the material parameters stored in particles.

6 Numerical damping

Real materials dissipates energy during movement. Temperature ans plastic deformation are common sources of energy dissipation. In numerical analysis numerical damping can be used for obtaining the quasi-static condition of the dynamic system.

7 Local damping

The numerical damping is a technique for getting a stationary solution of the dynamic system. In the MPM-Geomechanics simulator we have two type of numerical damping: the local (viscous) and the kinetic (dynamic relaxation) damping.

8 Quasi-static solution with local damping

The local damping is used to get a quasi-static solution of the dynamic system using a viscous nodal force. In each time step, a viscous force is applied in each node, whose magnitude is proportional and opposite to the nodal velocity.

$$f_{iI}^{dnplocal} = -\alpha |f_{iI}^{unb}| \hat{v}_{iI}$$

, where the unbalanced nodal force is:

$$f_{iI}^{unb} = f_{iI}^{int} + f_{iI}^{ext}$$

Therefore, the resulting discrete form of the motion equation with viscous nodal damping is:

$$\dot{p}_{iI} = f_{iI}^{int} + f_{iI}^{ext} + f_{iI}^{dnplocal}$$

9 Explicit algorithm

Algorithm 2 Explicit MPM integration scheme

1: Compute nodal mass: $m_I^k = \sum_{p=1}^{n_p} m_p N_{Ip}^k$
2: Compute nodal momentum: $p_{iI}^{k-1/2} = \sum_{p=1}^{n_p} m_p v_{ip}^{k-1/2} N_{Ip}^k$
3: Update seismic velocities from record (if required):
$a_i^{s,\,k-\frac{1}{2}} = a_i^{s,\,\text{record}}$
$v_i^{s,\,k+\frac{1}{2}} = \sum_{l=0}^{l=1-\frac{1}{2}} v_i^{s,\,l} + a_i^{s,\,\text{record}}\,\Delta t$
4: Impose nodal momentum boundary conditions: $p_{iI}^{k-1/2} = 0$ (fixed condition)
5: Compute internal forces: $f_{iI}^{intk} = -\sum_{p=1}^{n_p} \sigma_{ij}^{p\,k} S_{Ip,j}^k V_p$
6: Compute external forces: $f_{iI}^{extk} = \sum_{p=1}^{n_p} m_p b_i S_{Ip}^k + \int_\Gamma t_i N_I(x_i) dA$
7: Compute nodal unbalanced forces: $f_{iI}^{unbk} = f_{iI}^{intk} + f_{iI}^{extk}$
8: Compute damping forces (if required): $f_{iI}^{dnpk} = -\alpha |f_{iI}^{unbk}| \hat{v}_{iI}^{k-1/2}$
9: Compute total nodal forces: $f_{iI}^{totk} = f_{iI}^{unbk} + f_{iI}^{dnpk}$
10: Impose nodal force boundary conditions: $f_{iI}^{totk} = 0$ (fixed condition)
11: Update nodal momenta: $p_{iI}^{k+\frac{1}{2}} = p_{iI}^{k-1/2} + f_{iI}^{totk} \Delta t$
12: Update particle velocities: $v_{ip}^{k+\frac{1}{2}} = v_{ip}^{k-1/2} + \sum_{I=1}^{n_n} \frac{f_{iI}^{totk}}{m_I^k} N_{Ip}^k \Delta t$
13: Seismic nodal velocity: $v_{iI}^{k+\frac{1}{2}} = v_i^{s,\,k+\frac{1}{2}}$
14: Contact correction (if required): $v_{ip}^{k+\frac{1}{2}} = STLContactCorrection(v_{ip}^{k+\frac{1}{2}})$
15: Update particle positions: $x_{ip}^{k+1} = x_{ip}^k + \sum_{I=1}^{n_n} v_{iI}^{k+\frac{1}{2}} N_{Ip}^k \Delta t$
16: For MUSL scheme, recalculate nodal momentum: $p_{iI}^{k+1/2} = \sum_{p=1}^{n_p} m_p v_{ip}^{k+1/2} N_{Ip}^k$
17: Impose nodal momentum boundary conditions: $p_{iI}^{k+1/2} = 0$ (fixed condition)
18: Calculate nodal velocities: $v_{iI}^{k+1/2} = p_{iI}^{k+1/2} / m_I^k$
19: Compute particle strain increment: $\Delta\varepsilon_{ij}^{p\,k+1/2} = \frac{1}{2}\left(N_{p,j}^k v_i^{k-1/2} + N_{p,i}^k v_j^{k+1/2}\right)\Delta t$
20: Compute vorticity increment: $\Delta\omega_{ij}^{p\,k+1/2} = \frac{1}{2}\left(N_{p,j}^k v_i^{k+1/2} - N_{p,i}^k v_j^{k+1/2}\right)\Delta t$
21: Update particle density: $\rho_p^{k+1} = \rho_p^k / (1 + \text{tr}(\Delta\varepsilon_{ijp}^{k-1/2}))$
22: Update particle stress: $\sigma_{ij}^{p\,k+1} = StressUpdate(\sigma_{ij}^{p\,k}, \Delta\varepsilon_{ij}^{p\,k-1/2}, \Delta\omega_{ij}^{p\,k-1/2})$
23: $k \leftarrow k+1$

10 Project data structure

The directories with files and folders in this project are organized as follows:

- `src/ build/ qa/ external/ and inc/` sources, builds qa(tests and benchmarking), google-test folder and C++ headers files
- `manual/ docs/` program manual and documentation for doxygen
- `tests/` simulation results of analytical and numerical tests
- `examples/` application examples

11 Build commands

The following instructions outline the steps required to build the MPM-Geomechanics project.

11.1 Instruction commands using Linux/WSL or MSYS2 MINGW64 console line

After cloning the repository, navigate to the project directory and execute the following commands:

```
cd build/CMake
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j
```

11.2 Compilation in Windows

Prior to proceeding with these instructions, please consult the windows required programs section in this manual.

The simplest way to compile on windows is by using the '.bash' file at '/build/CMake' with 'MSYS2 MINGW64' console line, just execute:

```
cd project-directory/build/CMake
./cmake-build.bash
```

Alternatively, you can use the following commands:

```
cd project-directory/build/CMake
cmake -G "Unix Makefiles" -B build
cmake --build build
```

11.3 Compiling on Linux

Prior to proceeding please consult the Linux Required Programs section in this manual.

The simplest way to compile on Linux is by using the '.bash' file at '/build/CMake', just execute:

```
cd build/CMake
./cmake-build.bash
```

Alternatively, you can use the following commands:

```
cd build/CMake
cmake -G "Unix Makefiles" -B build
cmake --build build
```

12 Visual Studio Solution

For compiling the code in windows you can use the Visual Studio solution file '/build/MPM-Geomechanics.sln', and build it by pressing 'Ctr+B'. Alternatively you can compile it by using command in a *Developer Command Prompt*:

```
msbuild MPM-Geomechanics.sln -p:Configuration=Release
```

13 Make Compilation

For compile the code in a linux environment, execute the makefile inside the make folder: `MPM-Geomechanics\build\make\makefile`.

14 Program features

The MPM-Geomechanics program is an open source implementation of the Material Point Method (MPM) for geomechanics applications. It is designed to simulate the behavior of soils and rock under various loading conditions, including static and dynamic loads.

The main features of the program are:

- three dimensional formulation
- dynamic formulation
- shared memory parallelization using OpenMP
- body-based (pre-defined bodies, particles-list)
- constitutive models for soil and rock materials
- softening/hardening models to represent weakness during large deformations
- frictional contact algorithm using STL surfaces
- coupled fluid-mechanical formulation (*under development*)
- seismic boundary conditions

15 Compiled binaries

The recommended way to get the compiled binaries is by downloading the artifacts from the latest GitHub Actions workflow run.

- go to the [Actions page](https://github.com/fabricix/MPM-Geomechanics/actions).
- elect the latest run of the **MSBuild** workflow for Window, or **CI** for Linux.
- at the bottom, you will find the available artifacts under the **Artifacts** section.
- download the 'compiled-binaries' artifact to get the compiled code.

16 Required Programs

The following instructions outline the steps required to install all necessary programs to build the MPM-Geomechanics project and execute its test suite.

16.1 Required Programs - Windows

For Windows installation, the required programs can be installed using Winget and MSYS2. The following table summarizes the required programs, their installation methods, and the corresponding commands.

Program	Installation	Command
Winget	Microsoft's official website	Native of Windows
Git	via Winget	`winget install -e -id Git.Git -e -source winget`
MSYS2	via Winget	`winget install -e -id MSYS2.MSYS2 -source winget`
GitHub CLI	via MSYS2 MINGW64	`pacman -S mingw-w64-x86_64-github-cli`
Python	via MSYS2 MINGW64	`pacman -S mingw-w64-x86_64-python`
CMake	via MSYS2 MINGW64	`pacman -S mingw-w64-x86_64-cmake`
GCC	via MSYS2 MINGW64	`pacman -S mingw-w64-x86_64-gcc`
G++	via MSYS2 MINGW64	`pacman -S mingw-w64-x86_64-gcc`
Make	via MSYS2 MINGW64	`pacman -S make`

Table 1: Programs required for Windows installation.

Pre-requisite: Verify Winget installation

Make sure you have Winget installed. You can verify this by running `winget -version`. If you don't have Winget installed, you can get it from Microsoft's official website.

Step 1: Install Git, GitHub, and MSYS2

Run the following commands:

```
winget install -e --id Git.Git -e --source winget
winget install -e --id MSYS2.MSYS2 --source winget
```

Step 2: Install required packages in MSYS2 MINGW64

```
pacman -S mingw-w64-x86_64-github-cli
pacman -S mingw-w64-x86_64-python
pacman -S mingw-w64-x86_64-cmake
pacman -S mingw-w64-x86_64-gcc
pacman -S make
```

Step 3: Verify the installations

```
git --version
gh --version
python --version
cmake --version
make --version
gcc --version
g++ --version
```

16.2 Required Programs - Linux

For Linux installation, the required programs can be installed using the apt package manager. The following table summarizes the required programs, their installation methods, and the corresponding commands.

Program	Installation	Command
Git	via apt	`sudo apt install git`
GitHub CLI	via apt	`sudo apt install gh`
Python	via apt	`sudo apt install python3 python3-pip`
CMake	via apt	`sudo apt install cmake`
GCC	via apt	`sudo apt install build-essential`
G++	via apt	`sudo apt install build-essential`
Make	via apt	`sudo apt install build-essential`

Table 2: Programs required for Linux installation.

Step 1: Install all packages via apt

Listing 1: Installing required packages using *apt*

```
sudo apt install git
sudo apt install gh
sudo apt install python3 python3-pip
sudo apt install cmake
sudo apt install build-essential
```

Step 2: Verify the installations

```
git --version
gh --version
python3 --version
cmake --version
make --version
gcc --version
g++ --version
```

17 Code documentation

Code documentation is generated using Doxygen documentation generator. To generate the documentation run the following command:

```
doxygen Doxyfile
```

The HTML generated documentation is located in ‘/docs/index.html‘.

18 Execution

In order to run simulations in several terminal, you can add the compiled code in the system ‘PATH‘.

For Windows, you can follow these steps:

- Open the Start Search, type in "env", and select "Edit the system environment variables"

- In the System Properties window, click on the "Environment Variables..." button.
- In the Environment Variables window, under the "System variables" section, find and select the "Path" variable, then click on the "Edit..." button.
- In the Edit Environment Variable window, click on the "New" button and add the path to the folder where the compiled binary is located (e.g., `C:\path\to\MPM-Geomechanics\build\CMake\build\Re`
- Click "OK" to close all windows.

For Linux, you can add the following line to your `~/.bashrc` or `~/.bash_profile` file:

```
export PATH=$PATH:/path/to/MPM-Geomechanics/build/CMake/build
```

For both Windows and Linux, after adding the path, you can open a new terminal and run the program from any location by simply typing ‘MPM-Geomechanics‘ followed by the input file name.

19 Testing Compilation and Benchmarking

In order to compile the testing and benchmarking executables, please follow the instructions below.

19.1 How to Compile

Prior to proceeding with these instructions, please consult required programs in manual.

The tests use **GoogleTest**. It is necessary to import this library by cloning the official repository into the folder '/external'. Each developer must clone this repository independently.

```
cd external
git clone https://github.com/google/googletest.git
```

The simplest way to compile on Windows and Linux is by using the '.bash' file at '/build/qa' with 'MSYS2 MINGW64' console line, simply execute the following command in the directory 'MPM-Geomechanics/build/qa':

```
./cmake-build.bash
```

Alternatively, you can use the following commands:

```
cmake -G "Unix Makefiles" -B build
cmake --build build
```

These commands will generate two executables: 'MPM-Geomechanics-tests' and 'MPM-Geomechanics-benchmark'.

- 'MPM-Geomechanics-tests': Run testing using GoogleTest. All files ending with '.test.cpp' are testing files, you can find them in the directory 'qa/tests'.

- 'MPM-Geomechanics-benchmark': Run benchmark using GoogleTest. All files ending with '.benchmark.cpp' are performance files. You can find them in the directory 'qa/benchmark'.

19.2 How does benchmarking work?

If you are using Windows OS, make sure to use the MINGW64 command-line console.

19.3 Executable MPM-Geomechanics-benchmark

To run the benchmark correctly, a JSON file named `benchmark-configuration.json` is required. This file allows the user to specify values for each test. If the file does not exist or a value is missing, a default value will be used.

The executable `MPM-Geomechanics-benchmark` allows the following command-line arguments:

`<directory>`: Specifies which file should be used to run the benchmark. If no file is specified, the program will use `benchmark-configuration.json` located in the same directory as the executable. Example: `MPM-Geomechanics-benchmark configuration-file.json`

The executable `MPM-Geomechanics-benchmark` allows the following command-line flags:

`-log`: Shows more information about missing keys in the `benchmark-configuration.json` file

19.4 Script (executable): start-multi-benchmark.py

The performance test can also be executed using the 'start-multi-benchmark.py' script, which allows running benchmarks with one or more executables downloaded as artifacts from GitHub and stores the log results in separate folders. Each executable is automatically downloaded from GitHub as an artifact, using an ID specified in `start-multi-benchmark-configuration.json`.

Additionally, the benchmark configuration (number of martial points and number of threads) can be defined in the same file.

Example of a `start-multi-benchmark-configuration.json` file:

```
{
  "executables": {
    "win_benchmark_executable": "MPM-Geomechanics-benchmark.exe",  -> An executable located at build/qa/MPM-Geomechanics-benchmark.exe
    "win_github_id": "17847226555"  -> An artifact ID that represent it. can check out if it exists via gh run view <ID>
  },
  "parameters": {
    "particles": [ 15000, 25000, 50000],  -> material point to test
    "threads": [1, 5, 10]  -> number of threads to test
  }
}
```

This setup will run 3x3 (particles x threads) totaling 18 tests (3x3x2). The executable `start-multi-benchmark.py` allows the following command-line flags:

- `--clear`: Removes the **benchmark folder** (if it exists) before executing the performance tests.
- `--cache`: Uses the previously created cache instead of the `start-multi-benchmark-configuration.json` file.

20 Contact Documentation

In this section, we provide documentation and examples related to the contact algorithms based on the multi nodal velocity field implemented in MPM-Geomechanics.

21 Slave-master velocity contact approach

In the Slave-Master Contact method, there are two velocity fields, one for the master body v_{iI}^M and another for the slave body v_{iI}^S.

21.1 Contact detection

The contact between two bodies occurs when the following conditions are satisfied:

- The momenta of both bodies are mapped to the same grid node I;
- The normal velocities at the contact grid node I of the two bodies satisfy

$$\left(v_{iI}^M - v_{iI}^S\right) n_{iI}^M > 0. \tag{3}$$

21.2 Contact normal vector

The unit normal n_{iI}^M to the surface of body M at grid node I can be calculated from the mass gradient as

$$n_I = \frac{\sum_p m_p N_{Ip}}{\left|\sum_p m_p N_{Ip}\right|}. \tag{4}$$

This unit normal does not satisfy the collinearity condition $n_{iI}^M = -n_{iI}^S$, which leads to non-conservation of momentum and even penetration.

A collinear unit normal can be obtained by averaging the two unit normals, i.e.,

$$n_{iI}^{MS} = -n_{iI}^{SM} = \frac{n_{iI}^{M} - n_{iI}^{S}}{|n_{iI}^{M} - n_{iI}^{S}|}. \tag{5}$$

If body M is stiffer than body S, or if the surface of body M is flat/convex but the surface of body S is concave, choose the unit normal of body M as the collinear unit normal, i.e.,

$$n_{iI}^{MS} = -n_{iI}^{SM} = n_{iI}^{M}. \tag{6}$$

21.3 Contact Force

The contact force is obtained in a trial-correction approach. The momentum equation of each body is first integrated independently to obtain the trial solution as if both bodies were not in contact. If the trial solution satisfies the impenetrability condition, take the trial solution as the final true solution. If not, the contact force is applied at the contact grid nodes to prevent penetration.

The normal and tangential components of this force at the master body are defined by:

$$f_{iI}^{M,nor,k} = f_{jI}^{M,c,k} n_{jI}^{M,k} n_{iI}^{M,k}, \tag{7}$$

and

$$\min\left(\|f_{iI}^{M,tan,k}\|, \mu \|f_{iI}^{M,nor,k}\|\right) \frac{f_{iI}^{M,tan,k}}{\|f_{iI}^{M,tan,k}\|}, \tag{8}$$

respectively.

Here,

$$f_{iI}^{M,c,k} = \frac{1}{\left(m_I^{M,k} + m_I^{S,k}\right) \Delta t^k} \left(m_I^{M,k} \bar{p}_{iI}^{S,k+\frac{1}{2}} - m_I^{S,k} \bar{p}_{iI}^{M,k+\frac{1}{2}}\right), \tag{9}$$

represents the contact force for sticking contact.

21.4 Distance Correction

The previous contact condition may result in spurious contact. For example, when the space between two bodies approaching each other is less than twice the cell size, the previous conditions are satisfied at grid node I, identifying it as a contacted grid node, but the two bodies are not actually in contact at this time.

To avoid spurious contact, the detection condition can be improved by calculating the real distance between the two bodies.

Let X_I^M and X_I^S denote the position vectors emanating from grid node I to its closest particle in body M and body S, respectively. The distance between the two bodies can be calculated as the sum of the projections of these two position vectors onto the normal vectors of the two bodies at the grid node I:

$$D_I^{MS} = -X_I^M \cdot n_I^M - X_I^S \cdot n_I^S. \tag{10}$$

Thus, the contact detection condition can be modified as:

- The momenta of both bodies are mapped to the same grid node I;
- $\left(v_{iI}^M - v_{iI}^S\right) n_{iI}^M > 0$; and
- $D_I^{MS} \leq \lambda\, d_c$.

In which $\lambda\, d_c$ is used to take the particle size into account, d_c is the cell size, and λ is set to 0.5 by default, because two particles are used initially in each direction of a cell.

21.5 Implementation and Input File

The numerical implementation of the Slave-Master contact algorithm in MPM-Geomechanics follows the algorithm below:

Algorithm 3 MPM Multi-body Contact Algorithm

for each grid node i **do**
    Identify the bodies that have material points mapped to node i
    Reconstruct nodal mass m_i and momentum p_i, distinguishing contact nodes from regular nodes
    if node i is a grid contact node **then**
        Compute the external normal relative to each body
    end if
    Impose essential boundary conditions on grid nodal momentum
    Compute grid nodal internal, external, and total forces ($f^{\text{int}}, f^{\text{ext}}, f^{\text{tot}}$)
    Integrate the trial momentum of each body independently (without contact)
end for
for each grid contact node i **do**
    if body penetration is detected **then**
        Calculate the normal and tangential contact forces
        Correct the contact grid nodal momentum
    end if
end for
Update the position x_{ip} and velocity v_{ip} of each body material point

The Slave-Master contact algorithm is implemented in the MPM-Geomechanics code and can be activated by including the following lines in the input file:

```
"contact_manager": {
  "active": true,
  "real_distance_correction_coefficient": 0.5
}
```

The Slave-Master contact analysis settings are defined using the `"contact_manager"` keyword. In this context, the "active" keyword enables contact analysis, and `"real_distance_correction_coefficient"` defines the factor applied for contact correction, according to the theoretical documentation. If the value associated with is set to 0, `"real_distance_correction_coefficient"` the correction is deactivated.

The properties associated with each Slave-Master contact pair can be specified by including the following lines in the input file:

```
"contact":[
{
  "id": 1,
  "friction": 0.2,
  "master_id": 1,
  "slave_id": 2,
  "normal_type": "master"
}
]
```

The definition of Slave-Master contact properties for each body pair is made using the "contact" keyword. In this context, the "id" keyword defines the contact identifier, "friction" specifies the contact friction coefficient, `"master_id"` defines the master body identifier, `"slave_id"` defines

the slave body identifier, and `"normal_type"` specifies which normal will be used in the contact analysis.

22 Verification Problems

22.1 Block sliding problem - Single contact example

Introduction

This example models the sliding of a block on an inclined plane. The model was simulated six times, varying the friction coefficient μ (0.0, 0.1, 0.2, 0.3, 0.4, and 0.5) while keeping the inclination θ of the plane fixed at 45° and gravity g at $10\,\mathrm{m/s^2}$.

The results obtained were compared with the analytical solution of the problem, demonstrating convergence between the results. The purpose of this model is to verify the implementation of contact, as well as to demonstrate how to build a model for this type of analysis.

Analytical Solution

The analytical solution for the displacement of the block in the direction of the inclined plane with slope θ as a function of time, considering a coefficient of friction μ and the acceleration due to gravity g, is given by:

$$d = \frac{1}{2}\left(g\,\sin\theta - \mu\,g\,\cos\theta\right)t^2 \tag{11}$$

MPM Model

The MPM model consists of two bodies, one representing the inclined plane and the other the block, both created using the keyword `cuboid`. The first has dimensions $l_x = 32$, $l_y = 14$, and $l_z = 2$ m, with point 1 at $(0, 0, 0)$ and point 2 at $(32, 14, 2)$, while the second has dimensions $l_x = l_y = l_z = 10$ m, with point 1 at $(2, 2, 2)$ and point 2 at $(12, 12, 12)$.

For the elastic parameters:

$$E = 100 \times 10^6\,\mathrm{Pa}, \quad \rho = 2500\,\mathrm{kg/m^3}, \quad \nu = 0.3$$

The computational mesh has a cell size of $\Delta x = \Delta y = \Delta z = 1$ m. The inclination is simulated by tilting the gravitational acceleration, which is given by $(7.07, 0.0, -7.07)$ $\mathrm{m/s^2}$. The planes X_0, Y_0, Z_0, X_n, Y_n, and Z_n are defined as sliding, i.e., only tangential displacements are allowed.

Input File

```
{
    "time": 2.0,
    "critical_time_step_multiplier": 0.15,
    "gravity":[7.07,0.0,-7.07],
    "results":
    {
      "print": 25,
      "material_point_results": [ "id", "displacement", "material", "
         active", "pressure", "external_force", "contact", "velocity" ]
    },
    "mesh":
    {
      "cells_dimension":[ 1, 1, 1 ],
      "cells_number":[ 32, 14, 14 ],
      "origin":[ 0, 0, 0 ],
```

```
      "boundary_conditions":
      {
        "plane_X0": "sliding",
        "plane_Y0": "sliding",
        "plane_Z0": "sliding",
        "plane_Xn": "sliding",
        "plane_Yn": "sliding",
        "plane_Zn": "sliding"
      }
    },
    "material":[
      {
        "type":"elastic",
        "id":1,
        "young":100e6,
        "density":2500,
        "poisson":0.3
      },
      {
        "type":"elastic",
        "id":2,
        "young":100e6,
        "density":2500,
        "poisson":0.3
      }
    ],
    "body":
    [
      {
        "type":"cuboid",
        "id": 1,
        "point_p1":[ 0, 0, 0 ],
        "point_p2":[ 32, 14, 2 ],
        "material_id":1
      },
      {
        "type":"cuboid",
        "id": 2,
        "point_p1":[ 2, 2, 2 ],
        "point_p2":[ 12, 12, 12 ],
        "material_id":2
      }
    ],

    "contact_manager":{ "active": true },

    "contact":[
      {
        "id":1,
        "friction":0.5,
        "master_id":1,
        "slave_id":2,
        "normal_type":"master"
      }
    ]
  }
```

MPM Result Comparison

The MPM numerical results show good agreement with the analytical solution, with small deviations for the smaller values of friction coefficient.

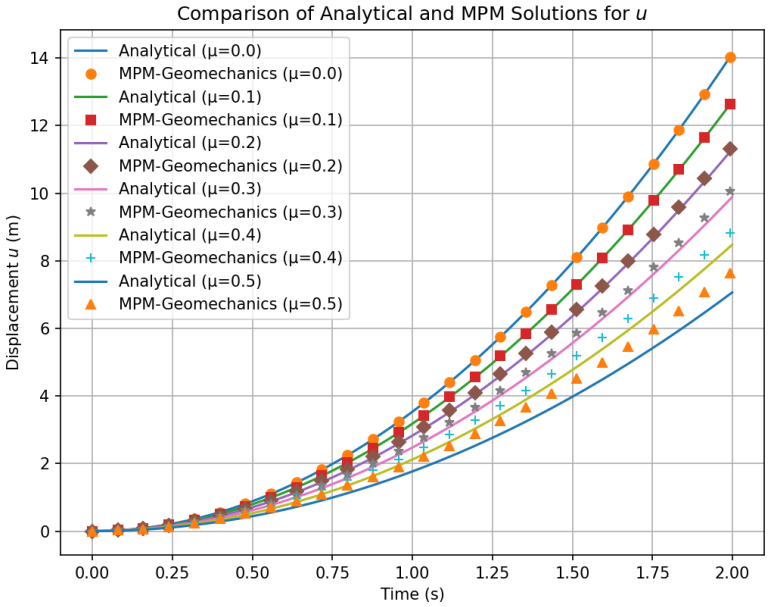


Figure 3: Verification of displacements obtained with MPM simulation.

22.2 Block sliding - Slave-Master multi contact example

Introduction

This example models the sliding of two blocks on an inclined plane.

The results obtained were compared with the analytical solution of the problem, demonstrating convergence between the results. The purpose of this model is to verify the implementation of multi contact, as well as to demonstrate how to build a model for this type of analysis.

Analytical Solution

The analytical solution for the displacement of each block in the direction of the inclined plane with slope θ as a function of time, considering a coefficient of friction μ and the acceleration due to gravity g, was shown the the singles contact verification examples.

MPM Model

The model was simulated keeping the inclination θ of the plane fixed at 45° and gravity g at $10\,\mathrm{m/s^2}$.

The MPM model consists of three bodies created using the keyword `cuboid`, one representing the inclined plane and the others representing two blocks. The first has dimensions $l_x = 32$, $l_y = 14$, and $l_z = 2$ m, with point 1 at $(0, 0, 0)$ and point 2 at $(32, 14, 2)$, while the second and the third have dimensions $l_x = l_z = 5$ and $l_y = 10$ m, with point 1 at $(2, 2, 2)$ and point 2 at $(7, 12, 7)$ for block one (wich body id equal to 2), and with point 1 at $(12, 2, 2)$ and point 2 at $(17, 12, 7)$, for block two (wich body id equal to 2). The friction coefficient μ was set as 0.2 for block one and as 0.0 for block two.

For the elastic parameters:

$$E = 100 \times 10^6\,\text{Pa}, \quad \rho = 2500\,\text{kg/m}^3, \quad \nu = 0.3$$

The computational mesh has a cell size of $\Delta x = \Delta y = \Delta z = 1$ m. The inclination of the plane fixed at 45° and gravity $g = 10\,\text{m/s}^2$ is simulated by tilting the gravitational acceleration, which, in the input file, is given by $(7.07, 0.0, -7.07)$ m/s^2. The planes X_0, Y_0, Z_0, X_n, Y_n, and Z_n are defined as sliding, i.e., only tangential displacements are allowed.

Input File

```
{
  "time": 2.0,
  "critical_time_step_multiplier": 0.15,
  "gravity":[7.07,0.0,-7.07],
  "results":
  {
    "print": 25,
    "material_point_results": [ "id", "displacement", "material", "
       active", "pressure", "external_force", "contact", "velocity" ]
  },
  "mesh":
  {
    "cells_dimension":[ 1, 1, 1 ],
    "cells_number":[ 32, 14, 14 ],
    "origin":[ 0, 0, 0 ],
    "boundary_conditions":
    {
      "plane_X0": "sliding",
      "plane_Y0": "sliding",
      "plane_Z0": "sliding",
      "plane_Xn": "sliding",
      "plane_Yn": "sliding",
      "plane_Zn": "sliding"
    }
  },
  "material":
  [
    {
      "type":"elastic",
      "id":1,
      "young":100e6,
      "density":2500,
      "poisson":0.3
    },
    {
      "type":"elastic",
      "id":2,
      "young":100e6,
      "density":2500,
      "poisson":0.3
    }
  ],
  "body":
  [
    {
      "type":"cuboid",
      "id": 1,
```

```
        "point_p1":[ 0, 0, 0 ],
        "point_p2":[ 32, 14, 2 ],
        "material_id":1
      },
      {
        "type":"cuboid",
        "id": 2,
        "point_p1":[ 2, 2, 2 ],
        "point_p2":[ 7, 12, 7 ],
        "material_id":2
      },
      {
        "type": "cuboid",
        "id": 3,
        "point_p1": [ 12, 2, 2 ],
        "point_p2": [ 17, 12, 7 ],
        "material_id": 2
      }
    ],
    "contact_manager":{
      "active": true
    },

    "contact":[
      {
        "id": 1,
        "friction": 0.2,
        "master_id": 1,
        "slave_id": 2,
        "normal_type": "master"
      },
      {
        "id": 2,
        "friction": 0.0,
        "master_id": 1,
        "slave_id": 3,
        "normal_type": "master"
      }
    ]
  }
```

MPM Result Comparison

The MPM numerical results show good agreement with the analytical solution for the displacement of both blocks.

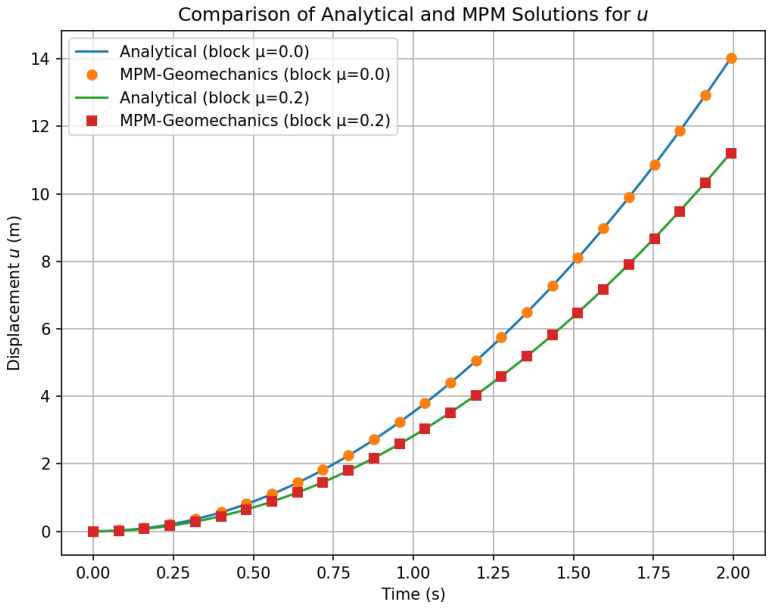


Figure 4: Verification of displacements obtained with MPM simulation.

22.3 Block falling - Contact real distance correction

Introduction

This example models two blocks falling with two different inicial velocities, on a fixed layer, and disregarding the effect of gravity. The aim of this example is to demonstrate the impact of the real distance correction for contact analysis.

MPM Model

The MPM model consists of three bodies created using the keyword `cuboid`, one representing a fixed layer and the others representing two blocks. The first has dimensions $l_x = 32$, $l_y = 14$, and $l_z = 2$ m, with point 1 at $(0, 0, 0)$ and point 2 at $(32, 14, 2)$, while the second and the third have dimensions $l_x = l_z = 5$ and $l_y = 10$ m, with point 1 at $(2, 2, 8)$ and point 2 at $(7, 12, 13)$, and initial velocity equal to four m/s, for block one (wich body id equal to 2), and with point 1 at $(12, 2, 4)$ and point 2 at $(17, 12, 9)$, for block two (wich body id equal to 2), and initial velocity equal to two m/s.

$$E = 100 \times 10^6\,\text{Pa}, \quad \rho = 2500\,\text{kg/m}^3, \quad \nu = 0.3$$

The computational mesh has a cell size of $\Delta x = \Delta y = \Delta z = 1$ m. The planes X_0, Y_0, Z_0, X_n, Y_n, and Z_n are defined as sliding, i.e., only tangential displacements are allowed.

Input File

```
{
  "time": 3.0,
  "critical_time_step_multiplier": 0.5,
  "gravity":[0.0, 0.0, 0.0],
  "results":
  {
    "print": 25,
    "material_point_results": [ "id", "displacement", "material", "
       active", "pressure", "external_force", "contact", "velocity" ]
```

```
    },
    "mesh":
    {
      "cells_dimension":[ 1, 1, 1 ],
      "cells_number":[ 32, 14, 14 ],
      "origin":[ 0, 0, 0 ],
      "boundary_conditions":
      {
        "plane_X0": "sliding",
        "plane_Y0": "sliding",
        "plane_Z0": "sliding",
        "plane_Xn": "sliding",
        "plane_Yn": "sliding",
        "plane_Zn": "sliding"
      }
    },
    "material":
    [
      {
        "type":"elastic",
        "id":1,
        "young":150e6,
        "density":2500,
        "poisson":0.3
      },
      {
        "type":"elastic",
        "id":2,
        "young":100e6,
        "density":2500,
        "poisson":0.3
      }
    ],
    "body":
    [
      {
        "type":"cuboid",
        "id": 1,
        "point_p1":[ 0, 0, 0 ],
        "point_p2":[ 32, 14, 2 ],
        "material_id":1
      },
      {
        "type": "cuboid",
        "id": 2,
        "point_p1": [ 2, 2, 8 ],
        "point_p2": [ 7, 12, 13 ],
        "material_id": 2,
        "initial_velocity": [ 0, 0, -4 ]
      },
      {
        "type": "cuboid",
        "id": 3,
        "point_p1": [ 12, 2, 4 ],
        "point_p2": [ 17, 12, 9 ],
        "material_id": 2,
        "initial_velocity": [ 0, 0, -2 ]
      }
```

```
    ],
    "contact_manager":{
      "active": true,
      "real_distance_correction_coefficient": 0.5
    },

    "contact":[
      {
        "id": 1,
        "friction": 0.2,
        "master_id": 1,
        "slave_id": 2,
        "normal_type": "master"
      },
      {
        "id": 2,
        "friction": 0.0,
        "master_id": 1,
        "slave_id": 3,
        "normal_type": "master"
      }
    ]
  }
```

MPM Result Comparison

In this example, Bodies 2 and 3 are initially separated by 6 m and 2 m, respectively. In the absence of external forces, they move at constant velocities of 4 m/s and 2 m/s, making contact with Body 1 at t = 1.5 s and t = 1.0 s, respectively.

In a displacement-time graph, Bodies 2 and 3 exhibit linear behavior until contacting Body 1, at which point a trend shift occurs. Graph analysis reveals that omitting the real-distance correction led to contact being detected earlier than predicted by the analytical solution. Conversely, applying the correction yielded contact times that align closely with analytical predictions.

In a velocity-time graph, the bodies maintain a constant velocity until contact, after which their direction of motion reverses. Consistent with the displacement results, the velocity analysis when real-distance corretion is not applied indicates contact even when the gap between the bodies equals the mesh cell size.

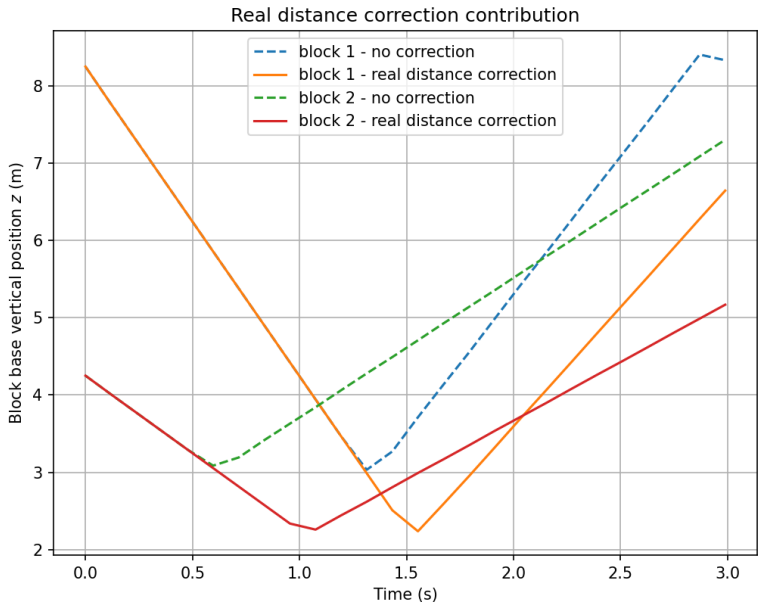


Figure 5: Displacement comparison.

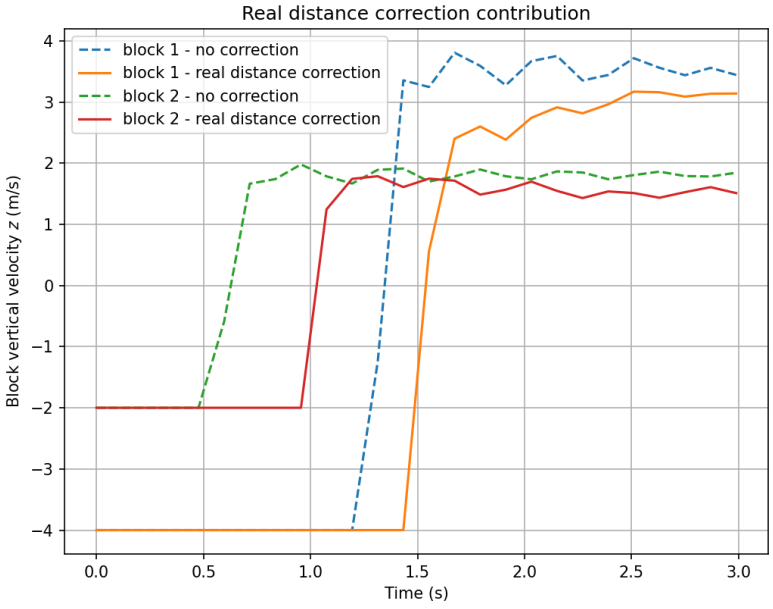


Figure 6: Velocity comparison.

22.4 Boussinesq's Problem

Introduction

In Geomechanics, the Boussinesq's problem refers to the point load acting on a surface of an elastic half-space. The boundary conditions for this problem are:

- The load P is applied only at one point, at the origin.

- The load is zero at any other point.
- For any point infinitely distant from the origin, the displacements must all vanish.

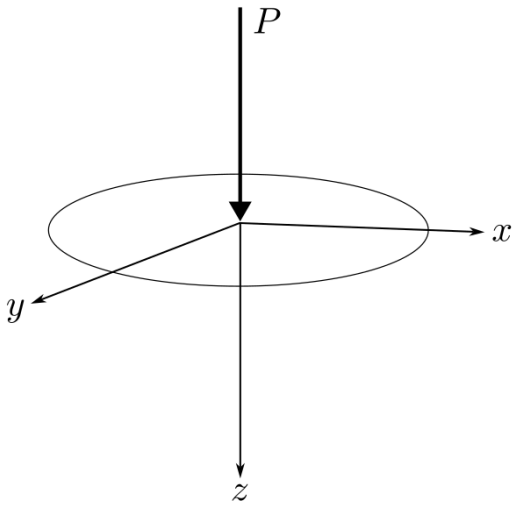


Figure 7: Boussinesq's problem.

Analytical Solution

The analytical solution of the vertical displacement field is:

$$u_z(x,y,z) = \frac{P}{4\pi G d}\left(2(1-\nu) + \frac{z^2}{d^2}\right)$$

where $G = \frac{E}{2(1+\nu)}$ is the shear modulus of the elastic material, ν is the Poisson ratio, and $d = \sqrt{x^2+y^2+z^2}$ is the total distance from the load to the point.

MPM Model and Result Comparison

To model the displacement field generated by the point load, we create an elastic body with dimensions $l_x = l_y = l_z = 1\,\mathrm{m}$, using the keyword `"cuboid"` with Point 1 at (0,0,0) and Point 2 at (1,1,1).

For the elastic parameters:

$$E = 200 \times 10^6\,\mathrm{Pa}, \quad \rho = 1500\,\mathrm{kg/m^3}, \quad \nu = 0.25$$

The computational mesh has cell dimensions $\Delta x = \Delta y = \Delta z = 0.1\,\mathrm{m}$. A nodal load of magnitude 1 is applied in the vertical direction at the midpoint of the upper surface. The plane Z_n is free, while all other planes are sliding (only tangential displacements allowed). Dynamic relaxation is used to reach a static solution, through the keyword `"damping"` with type `"kinetic"`.

Input File

```
{
  "body": {
    "elastic-cuboid-body": {
      "type":"cuboid",
```

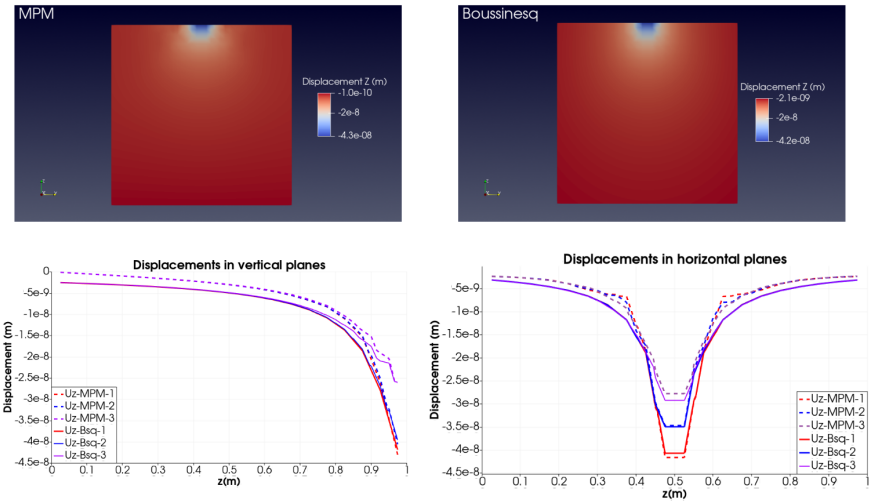


Figure 8: Comparison between analytical and MPM results.

```
        "id":1,
        "point_p1":[0.0,0.0,0],
        "point_p2":[1,1,1],
        "material_id":1
      }
    },
    "materials": {
      "material-1": {
        "type":"elastic",
        "id":1,
        "young":200e6,
        "density":1500,
        "poisson":0.25
      }
    },
    "mesh": {
      "cells_dimension":[0.1,0.1,0.1],
      "cells_number":[10,10,10],
      "origin":[0.0,0.0,0.0],
      "boundary_conditions": {"plane_Zn":"free"}
    },
    "time":0.025,
    "time_step_multiplier":0.3,
    "nodal_point_load": [[[0.5, 0.5, 1.0], [0.0, 0.0, -1.0]]],
    "damping": {"type":"kinetic" }
  }
```

The MPM numerical results show good agreement with the analytical solution, with small deviations due to discretization and the representation of the domain by particles with finite volume. Errors decrease with mesh refinement.

22.5 Base Acceleration Example

Introduction

In this example we model the motion of an elastic body subjected to a dynamic boundary condition. The body is a cuboid with dimensions $l_x = l_y = 0.3\,\text{m}$ and $l_z = 0.8\,\text{m}$, with lower coordinate point $p_{min} = (0.4, 0.4, 0.0)\,\text{m}$.

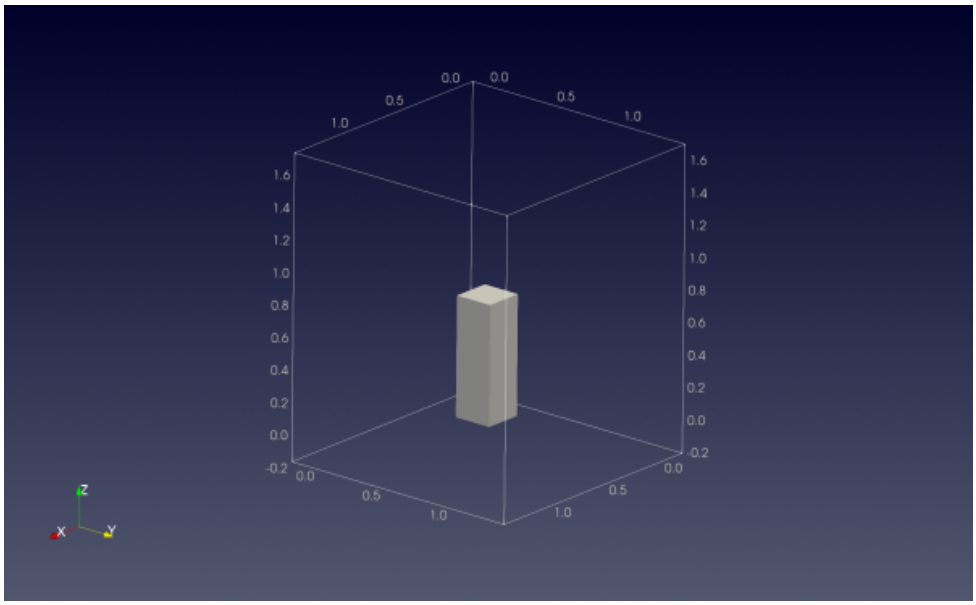


Figure 9: Geometry of the elastic body.

The base acceleration is defined as:

$$\ddot{u} = A\,2\pi f \cos(2\pi f t + \alpha)$$

The total simulation time is $T = 2\,\mathrm{s}$, with a time step $\Delta t = 10^{-4}\,\mathrm{s}$. Material properties:

$$\rho = 2500\,\mathrm{kg/m^3}, \quad E = 100 \times 10^6\,\mathrm{Pa}, \quad \nu = 0.25$$

MPM Model

The MPM model consists of uniformly distributed particles inside the body, created with the keywords `"body"` and `"cuboid"`. The mesh dimensions are $\Delta x = \Delta y = \Delta z = 0.1\,\mathrm{m}$, and it covers the full expected displacement range of the body. The mesh uses $n_x = 12, n_y = 12, n_z = 15$.

Input File

```
{
  "stress_scheme_update":"USL",
  "shape_function":"GIMP",
  "time":10,
  "time_step":0.0005,
  "gravity":[0.0,0.0,0.0],
  "n_threads":1,
  "damping": {
    "type":"local",
    "value":0.0
  },
  "results": {
    "print":100,
    "fields":["id","displacement","velocity","material","active","body
       "]
  },
  "n_phases":1,
  "mesh": {
```

```
      "cells_dimension":[0.1,0.1,0.1],
      "cells_number":[10,10,15],
      "origin":[0,0,0]
    },
    "earthquake": {
      "active": true,
      "file": "base_acceleration.csv",
      "header": true
    },
    "material": {
      "elastic_1": {
        "type":"elastic",
        "id":1,
        "young":10e6,
        "density":2500,
        "poisson":0.25
      }
    },
    "body": {
      "columns_1": {
        "type":"cuboid",
        "id":1,
        "point_p1":[0.2,0.2,0],
        "point_p2":[0.5,0.5,1.0],
        "material_id":1
      }
    }
  }
```

Earthquake Block Parameters

```
  "earthquake": {
    "active": true,
    "file": "base_acceleration.csv",
    "header": true
  }
```

Where:

- **active**: Enables or disables seismic loading.
- **file**: Path to the CSV file containing time, acceleration_x, acceleration_y, and acceleration_z.
- **header**: True if the CSV has a header row.

Example of the first lines of the record:

```
t,ax,ay,az
0.0,-1.8849555921538759,-0.9424777960769379,-0.0
5e-05,-1.8849554991350466,-0.9424777844495842,-0.0
0.0001,-1.884955220078568,-0.9424777495675233,-0.0
0.00015000000000000001,-1.8849547549844674,-0.9424776914307561,-0.0
...
```

Post-processing (results and visualizations)

Particle results are found in the `/particle` folder and grid results in the `/grid` folder located inside the current working directory. Particle results(`particle_1.vtu`, `particle_2.vtu`, ..., `particle_41.vtu`) are referenced in `particleTimeSerie.pvd`, which can be opened in ParaView: `File - Open - particleTimeSerie.pvd`The mesh can be loaded with `File - Open - eulerianGrid.vtu`

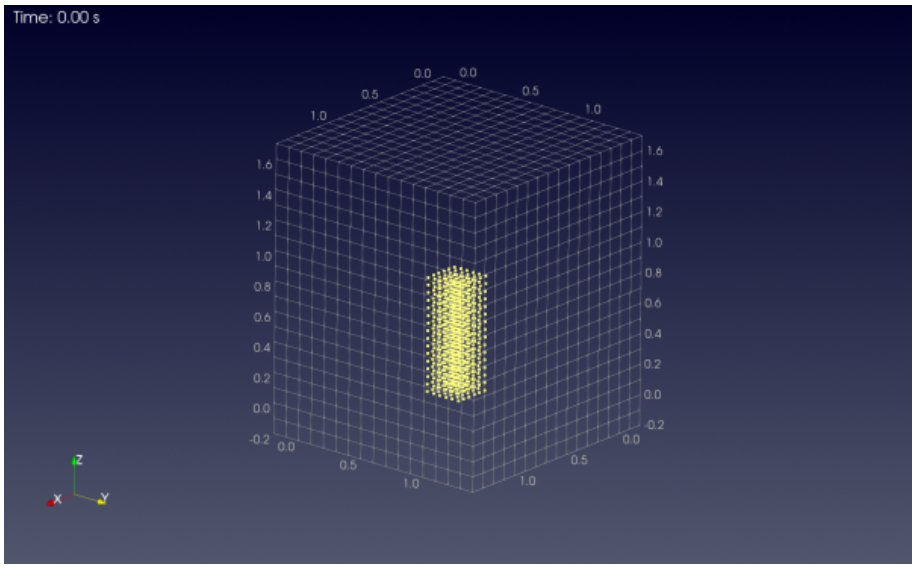


Figure 10: Particles and mesh of the analyzed case.

Verification of Dynamic Boundary Condition

The velocity from the analytical input function

$$\dot{u} = A\sin(2\pi ft + \alpha)$$

was compared with the particle velocity at the base of the model. The results show excellent agreement between analytical and MPM-calculated velocities.

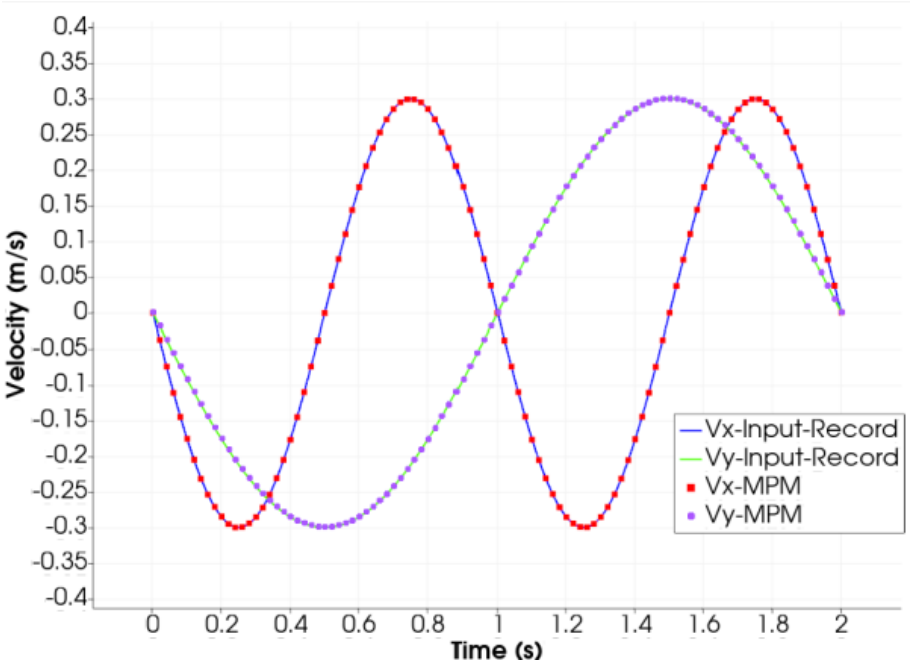


Figure 11: Verification of velocities obtained with MPM simulation.

22.6 Undrained strength ratio

In this example a 2D slope model is configured to setting up an undrained strength ratio of $S_u/\sigma_v = 0.001$. First, an initial simulation is performed with elastic parameters and kinetic damping for steady state condition, and then the stress field was saved with the command `"save_state":true`. Secondly, the stress field is loaded by `"load_state":true` and the undrained strength ratio is setting up using `"su_vertical_stress":0.001`.

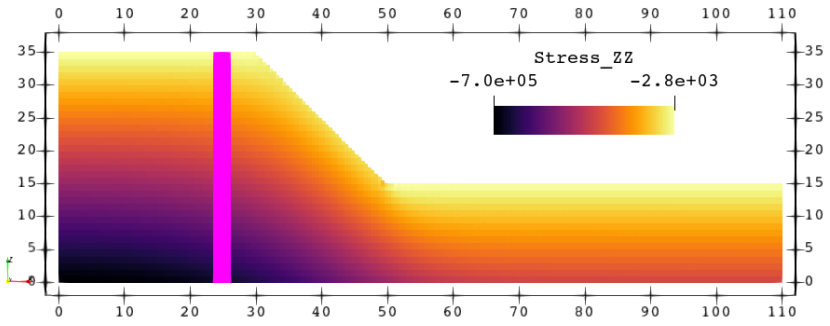


Figure 12: Elastic stress field

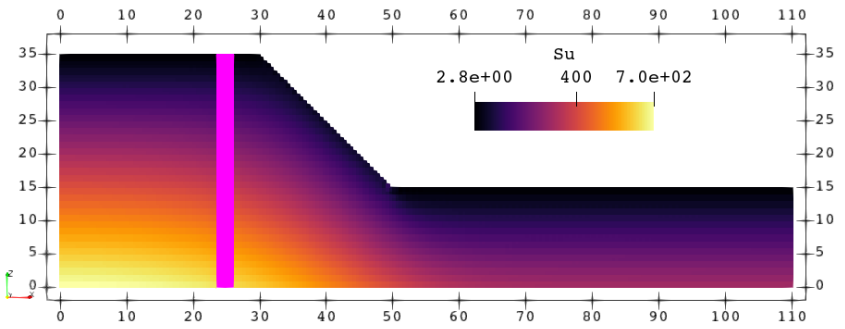


Figure 13: Su factor obtained from initial stress field

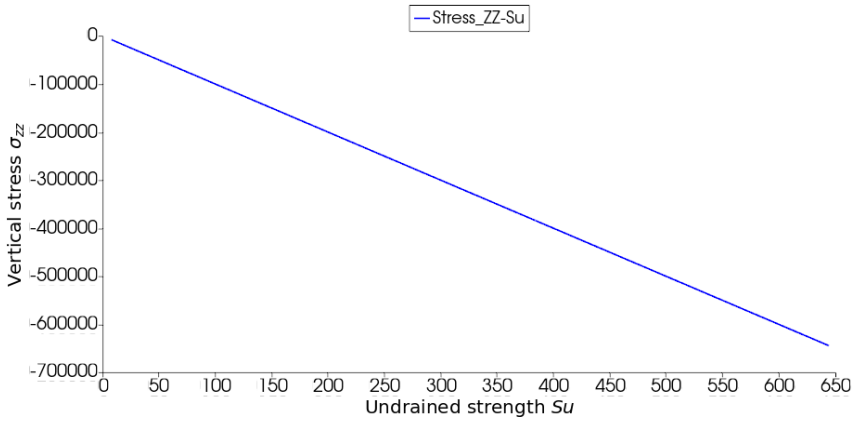


Figure 14: Stress Su relation along the vertical plane defined by colored particles